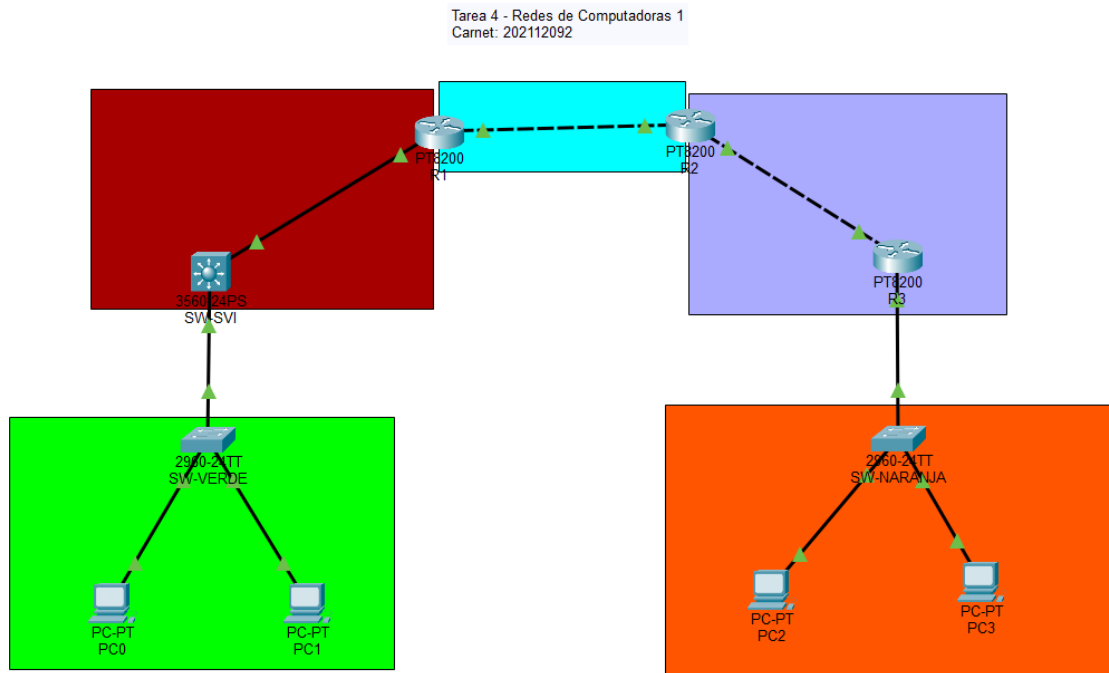


TOPOLOGIA COMPLETA



SW-VERDE

```
SW-VERDE>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	VLAN_VERDE_1	active	Fa0/2
20	VLAN_VERDE_2	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-VERDE>show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Fa0/1	10,20			
Port	Vlans allowed and active in management domain			
Fa0/1	10,20			
Port	Vlans in spanning tree forwarding state and not pruned			
Fa0/1	10,20			

SW-SVI

```
SW-SVI
Physical Config CLI Attributes
IOS Command Line Interface

SW-SVI#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 3 subnets
C       10.0.1.0 is directly connected, FastEthernet0/2
R       10.0.2.0 [120/1] via 10.0.1.2, 00:00:05, FastEthernet0/2
R       10.0.3.0 [120/2] via 10.0.1.2, 00:00:05, FastEthernet0/2
    192.168.92.0/25 is subnetted, 2 subnets
C       192.168.92.0 is directly connected, Vlan10
C       192.168.92.128 is directly connected, Vlan20
R       192.168.93.0/25 is subnetted, 2 subnets
R       192.168.93.0 [120/2] via 10.0.1.2, 00:00:05, FastEthernet0/2
R       192.168.93.128 [120/2] via 10.0.1.2, 00:00:05, FastEthernet0/2

SW-SVI#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 9 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
FastEthernet0/2      22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  192.168.92.0
Passive Interface(s):
  Vlan10
  Vlan20
Routing Information Sources:
  Gateway         Distance      Last Update
  10.0.1.2         120           00:00:10
Distance: (default is 120)
SW-SVI#
```

```
SW-SVI
Physical Config CLI Attributes
IOS Command Line Interface

SW-SVI#show vlan brief

VLAN Name                Status Ports
-----
1    default                active Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                   Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                   Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                   Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                   Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                   Fa0/24, Gig0/1, Gig0/2
10   VLAN_VERDE_1            active
20   VLAN_VERDE_2            active Fa0/3
1002 fddi-default           active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active

SW-SVI#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20

Port      Vlans allowed and active in management domain
Fa0/1     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20

SW-SVI#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 3 subnets
C       10.0.1.0 is directly connected, FastEthernet0/2
R       10.0.2.0 [120/1] via 10.0.1.2, 00:00:05, FastEthernet0/2
R       10.0.3.0 [120/2] via 10.0.1.2, 00:00:05, FastEthernet0/2
    192.168.92.0/25 is subnetted, 2 subnets
```

SW-NARANJA

```
SW-NARANJA>show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
30 VLAN_NARANJA_1	active	Fa0/2
40 VLAN_NARANJA_2	active	Fa0/3
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW-NARANJA>show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	30,40

Port	Vlans allowed and active in management domain
Fa0/1	30,40

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	30,40

ROUTER 1

```
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	10.0.1.2	YES	manual	up	up
GigabitEthernet0/0/1	10.0.2.1	YES	manual	up	up
GigabitEthernet0/0/2	unassigned	YES	unset	down	down
GigabitEthernet0/0/3	unassigned	YES	unset	down	down

```
Router#show ip route
```

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.0.1.0/30 is directly connected, GigabitEthernet0/0/0
L    10.0.1.2/32 is directly connected, GigabitEthernet0/0/0
C    10.0.2.0/30 is directly connected, GigabitEthernet0/0/1
L    10.0.2.1/32 is directly connected, GigabitEthernet0/0/1
O E2  10.0.3.0/30 [110/20] via 10.0.2.2, 00:18:21, GigabitEthernet0/0/1
192.168.92.0/25 is subnetted, 2 subnets
R    192.168.92.0 [120/1] via 10.0.1.1, 00:00:13, GigabitEthernet0/0/0
R    192.168.92.128 [120/1] via 10.0.1.1, 00:00:13, GigabitEthernet0/0/0
192.168.93.0/25 is subnetted, 2 subnets
O E2  192.168.93.0 [110/20] via 10.0.2.2, 00:18:21, GigabitEthernet0/0/1
O E2  192.168.93.128 [110/20] via 10.0.2.2, 00:18:21, GigabitEthernet0/0/1
```

```
Router#show ip protocols
```

```
% Invalid input detected at '^' marker.
```

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.3.1	1	FULL/DR	00:00:38	10.0.2.2	GigabitEthernet0/0/1

ROUTER 2

```
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  10.0.2.2        YES manual up          up
GigabitEthernet0/0/1  10.0.3.1        YES manual up          up
GigabitEthernet0/0/2  unassigned      YES unset  down        down
GigabitEthernet0/0/3  unassigned      YES unset  down        down
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O E2   10.0.1.0/30 [110/20] via 10.0.2.1, 00:21:49, GigabitEthernet0/0/0
C      10.0.2.0/30 is directly connected, GigabitEthernet0/0/0
L      10.0.2.2/32 is directly connected, GigabitEthernet0/0/0
C      10.0.3.0/30 is directly connected, GigabitEthernet0/0/1
L      10.0.3.1/32 is directly connected, GigabitEthernet0/0/1
       192.168.92.0/25 is subnetted, 2 subnets
O E2   192.168.92.0 [110/20] via 10.0.2.1, 00:21:49, GigabitEthernet0/0/0
O E2   192.168.92.128 [110/20] via 10.0.2.1, 00:21:49, GigabitEthernet0/0/0
       192.168.93.0/25 is subnetted, 2 subnets
D      192.168.93.0 [90/28416] via 10.0.3.2, 00:22:57, GigabitEthernet0/0/1
D      192.168.93.128 [90/28416] via 10.0.3.2, 00:22:57, GigabitEthernet0/0/1

Router>show ip protocols
^
% Invalid input detected at '^' marker.

Router>show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address         Interface
10.0.2.1         1     FULL/BDP        00:00:38    10.0.2.1        GigabitEthernet0/0/0
Router>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address      Interface       Hold Uptime    SRTT  RTO  Q  Seq
                   (sec)          (ms)          Cnt  Num
0   10.0.3.2       Gig0/0/1        10   00:22:58     40   1000  0   9
```

ROUTER 3

```
R3>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  10.0.3.2        YES manual up          up
GigabitEthernet0/0/1  unassigned      YES unset up          up
GigabitEthernet0/0/1.30 192.168.93.1    YES manual up          up
GigabitEthernet0/0/1.40 192.168.93.129  YES manual up          up
GigabitEthernet0/0/2  unassigned      YES unset down        down
GigabitEthernet0/0/3  unassigned      YES unset down        down
R3>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

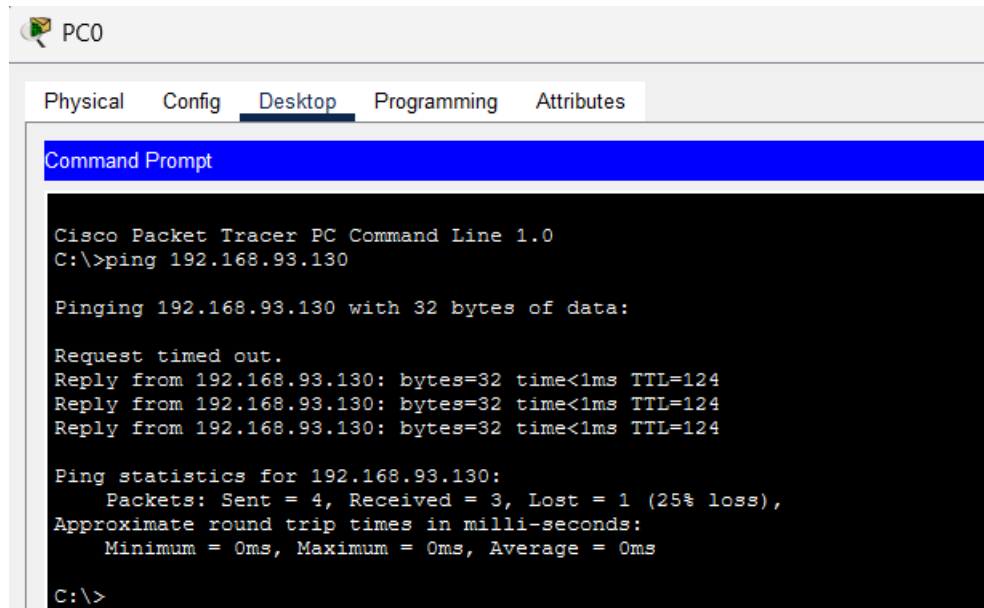
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
D EX   10.0.1.0/30 [170/281856] via 10.0.3.1, 00:20:12, GigabitEthernet0/0/0
D EX   10.0.2.0/30 [170/281856] via 10.0.3.1, 00:20:12, GigabitEthernet0/0/0
C      10.0.3.0/30 is directly connected, GigabitEthernet0/0/0
L      10.0.3.2/32 is directly connected, GigabitEthernet0/0/0
       192.168.92.0/25 is subnetted, 2 subnets
D EX   192.168.92.0 [170/281856] via 10.0.3.1, 00:20:12, GigabitEthernet0/0/0
D EX   192.168.92.128 [170/281856] via 10.0.3.1, 00:20:12, GigabitEthernet0/0/0
       192.168.93.0/24 is variably subnetted, 4 subnets, 2 masks
C      192.168.93.0/25 is directly connected, GigabitEthernet0/0/1.30
L      192.168.93.1/32 is directly connected, GigabitEthernet0/0/1.30
C      192.168.93.128/25 is directly connected, GigabitEthernet0/0/1.40

R3>show ip protocols
^
% Invalid input detected at '^' marker.

R3>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address      Interface       Hold Uptime    SRTT  RTO  Q  Seq
                   (sec)          (ms)          Cnt  Num
0   10.0.3.1       Gig0/0/0        12   00:22:20     40   1000  0   21
```

PING PC0 - PC3



The screenshot shows the Cisco Packet Tracer interface for PC0. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the output of a ping command to 192.168.93.130. The output shows a request timed out, followed by three successful replies from 192.168.93.130 with 32 bytes of data, a time of less than 1ms, and a TTL of 124. The ping statistics for 192.168.93.130 are also displayed, showing 4 packets sent, 3 received, and 1 lost (25% loss). The approximate round trip times in milliseconds are shown as Minimum = 0ms, Maximum = 0ms, and Average = 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.93.130

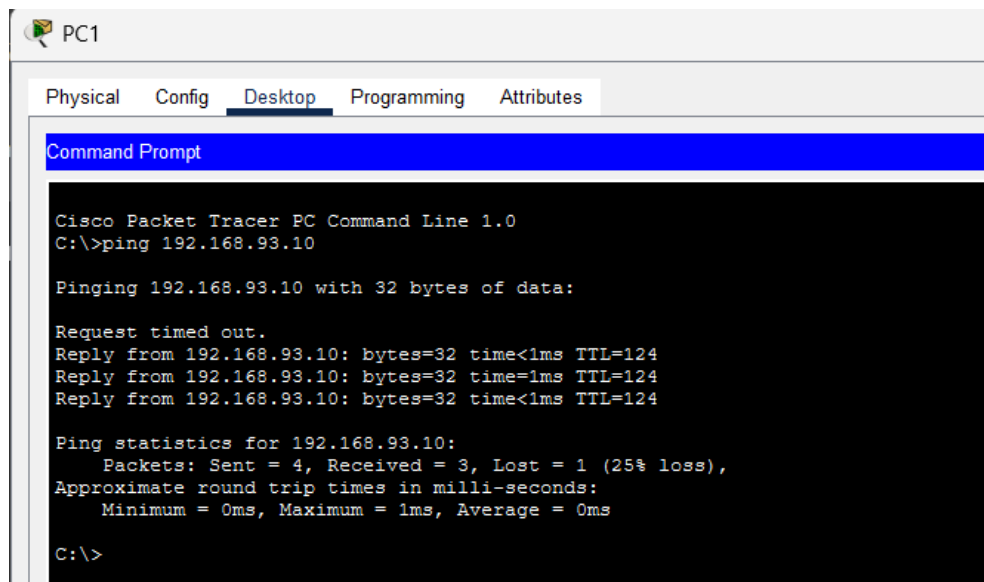
Pinging 192.168.93.130 with 32 bytes of data:

Request timed out.
Reply from 192.168.93.130: bytes=32 time<1ms TTL=124
Reply from 192.168.93.130: bytes=32 time<1ms TTL=124
Reply from 192.168.93.130: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.93.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PING PC1 - PC2



The screenshot shows the Cisco Packet Tracer interface for PC1. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the output of a ping command to 192.168.93.10. The output shows a request timed out, followed by three successful replies from 192.168.93.10 with 32 bytes of data, a time of less than 1ms, and a TTL of 124. The ping statistics for 192.168.93.10 are also displayed, showing 4 packets sent, 3 received, and 1 lost (25% loss). The approximate round trip times in milliseconds are shown as Minimum = 0ms, Maximum = 1ms, and Average = 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.93.10

Pinging 192.168.93.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.93.10: bytes=32 time<1ms TTL=124
Reply from 192.168.93.10: bytes=32 time<1ms TTL=124
Reply from 192.168.93.10: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.93.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PING PC0 – PC2

```
C:\>ping 192.168.93.10

Pinging 192.168.93.10 with 32 bytes of data:

Reply from 192.168.93.10: bytes=32 time<1ms TTL=124
Reply from 192.168.93.10: bytes=32 time<1ms TTL=124
Reply from 192.168.93.10: bytes=32 time=9ms TTL=124
Reply from 192.168.93.10: bytes=32 time=7ms TTL=124

Ping statistics for 192.168.93.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 4ms

C:\>|
```

REFLEXION

¿Qué aprendí y cómo enfrenté los errores?

Durante el desarrollo de esta tarea aprendí a configurar diferentes métodos de comunicación entre VLANs, utilizando SVI en un switch de capa 3 y Router-on-a-Stick mediante subinterfaces. También aprendí a utilizar RIP, OSPF y EIGRP dentro de una misma topología y a realizar redistribución de rutas para lograr comunicación entre redes que utilizan protocolos diferentes. Para solucionar los errores fui verificando cada etapa mediante comandos como show ip interface brief, show vlan brief, show interfaces trunk y show ip route, además de realizar pruebas de ping antes de continuar con la siguiente configuración.

Conclusión técnica

La práctica permitió comprobar que una red puede utilizar diferentes protocolos de enrutamiento y mantener comunicación completa mediante la redistribución de rutas. También se comprobó la importancia de configurar correctamente las VLANs, enlaces troncales, gateways e interfaces de capa 3. La verificación progresiva facilitó identificar errores de puertos, interfaces y configuración antes de implementar el enrutamiento completo.